Department of Computer Engineering
CMPE326, Spring 2025-26
Homework 1: Minimum Spanning Tree

1. Introduction

A spanning tree is defined as a tree-like subgraph of a connected, undirected graph that includes all the vertices of the graph. It is a subset of the edges of the graph that forms a tree (acyclic) where every node of the graph is a part of the tree. The **minimum spanning tree** has all the properties of a spanning tree with an added constraint of having the minimum possible weights among all possible spanning trees. Like a spanning tree, there can also be many possible MSTs for a graph.

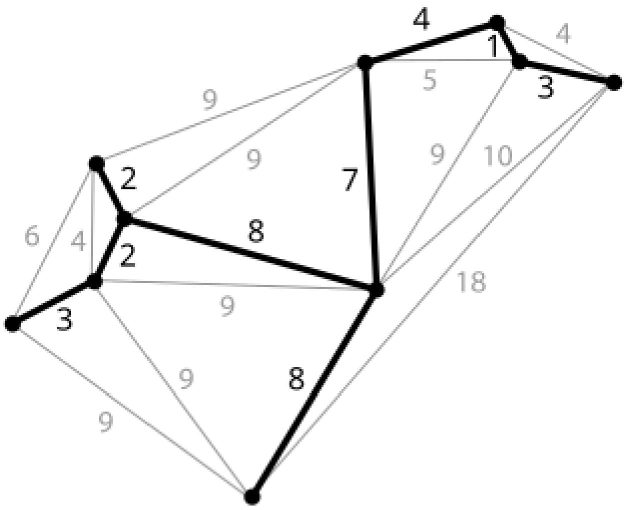

Figure: A unweighted graph and its minimum spanning tree.

2. Assignment

Given an un-directed complete graph represented using adjacency matrix where each entry represents the weight of the edges. You need to write Python code to find a minimum spanning tree (MST) for the complete graph and report the minimum spanning weight and the tree. You are allowed to use any standard MST algorithms (either Prim's or Kruskal's MST or any easy Algorithm). You are allowed to use set, dictionary of the python library.

The adjacency matrix of the un-directed graph will be given as 'udg.dat' file. Your program should read it from current working directory. You should display the minimum spanning weight on the standard output and save the minimum spanning tree in the current working directory with the name 'mst.dat'.

3. Submission

- Each student must submit his or her own work.
- You must implement using Python programing only and test your program with your own file in addition to the given‘udg.dat’ file uploaded on LMS.
- In the evaluation, additional hidden test cases are going to be used. Thus, only passing the sample test case does not mean that you are going to take full grade.
- Your works are going to be evaluated with Python3 interpreter.
- A similarity check is going to be applied to your submissions. In case of any plagiarism detection, you are going to get 0 point.
- Your program must read the input file ‘udg.dat’ from current working directory. You should display the minimum spanning weight on the standard output and save the minimum spanning tree in the current working directory with the name ‘mst.dat’. The submissions that use any other reading or writing strategy are not going to be evaluated.
- You may be asked for a demo session.
- The final submission must be as a python project saved in folder ‘hw1’ containing the main.py and other necessary items.